

Day 3: DNS Server ConfigurationTheory

Task 1 — Install and Configure a DNS Server

Configuration Used

- DNS server platform: BIND on Linux.
- DNS server IP address: 192.168.1.10.
- DNS zone: musicmania.in.
- Zone type: Primary/authoritative forward lookup zone.

BIND Zone Configuration

- Create the zone definition for musicmania.in in the BIND configuration.
- Create the zone file for musicmania.in and define the SOA and NS records.
- Set the DNS server as the authoritative name server for the zone.

Example Zone File

```
$TTL 86400
@      IN      SOA      ns1.musicmania.in. admin.musicmania.in. (
                                2026081401
                                3600
                                900
                                604800
                                86400 )
@      IN      NS       ns1.musicmania.in.
ns1    IN      A        192.168.1.10
```

Service Verification

```
sudo named-checkconf
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
sudo systemctl restart bind9
sudo systemctl status bind9
```

- The configuration and zone file are checked before restarting the DNS service.
- A successful service status confirms that BIND is running with the configured zone.

Task 2 — Add an A Record and Verify DNS Resolution

A Record Added

- Hostname: www.musicmania.in.
- Record type: A.
- IPv4 address: 192.168.1.50.

Zone Record

```
www      IN      A       192.168.1.50
```

Verification with nslookup

```
nslookup www.musicmania.in 192.168.1.10
```

Expected Result

- The DNS response should identify www.musicmania.in and return 192.168.1.50.
- This confirms that the authoritative DNS server is correctly mapping the hostname to the configured IPv4 address.

Alternative Verification with dig

```
dig @192.168.1.10 www.musicmania.in
```

Completed Result

- Query: www.musicmania.in
- Type: A
- Returned IPv4 address: 192.168.1.50
- DNS mapping: Working correctly.

Task 3 — Configure and Test a CNAME Record

CNAME Record Added

- Alias: play.musicmania.in.
- Canonical name: www.musicmania.in.
- Record type: CNAME.

Zone Record

```
play      IN      CNAME    www.musicmania.in.
```

How Resolution Works

- A DNS query for play.musicmania.in first follows the CNAME to www.musicmania.in.
- The A record for www.musicmania.in then supplies the IPv4 address 192.168.1.50.

Verification

```
nslookup play.musicmania.in 192.168.1.10
dig @192.168.1.10 play.musicmania.in
```

Expected Result

- play.musicmania.in resolves through the CNAME to www.musicmania.in.
- The final address returned is 192.168.1.50.

Completed Result

- play.musicmania.in → CNAME → www.musicmania.in → A → 192.168.1.50.
- The CNAME test succeeds because the alias ultimately resolves to the configured A record.

Task 4 — DNS Troubleshooting Procedure

Problem

- The DNS server is not resolving musicmania.in as expected.

Step 1 — Check the Configuration

```
sudo named-checkconf
```

- This checks the BIND configuration syntax.

Step 2 — Check the Zone File

```
sudo named-checkzone musicmania.in /etc/bind/db.musicmania.in
```

- This checks the zone file for syntax and record errors.

Step 3 — Check the DNS Service

```
sudo systemctl status bind9
```

- Confirm that the BIND service is active and running.

Step 4 — Restart the DNS Service

```
sudo systemctl restart bind9
```

- Restart BIND after correcting configuration or zone-file changes.

Step 5 — Test Local Resolution

```
dig @192.168.1.10 www.musicmania.in
```

- Confirm that the DNS server returns the expected A record of 192.168.1.50.

Step 6 — Test with nslookup

```
nslookup www.musicmania.in 192.168.1.10
```

- Verify that the hostname resolves correctly through the configured DNS server.

Step 7 — Check Logs if the Problem Continues

```
sudo journalctl -u bind9 --no-pager -n 50
```

- Review recent BIND service messages for configuration, zone-loading, or permission errors.

Common Problems to Check

- Incorrect zone name or zone-file path in the BIND configuration.
- Syntax errors in the SOA, NS, A, or CNAME records.
- Incorrect file permissions preventing BIND from reading the zone file.
- BIND service not running or failing to load the zone.
- Client configured to use a different DNS server instead of 192.168.1.10.

Final Verification

- After correcting the issue, run both nslookup and dig again and confirm that www.musicmania.in returns 192.168.1.50 and play.musicmania.in resolves through the CNAME.